

Bavesh Raam S

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Computer Science student specializing in Artificial Intelligence with hands-on experience in deep learning, embedded systems, and full-stack development. Focused on building deployable, production-grade systems that solve real-world problems.

Education

B.Tech in Computer Science and Engineering (Artificial Intelligence)

Amrita Vishwa Vidyapeetham, Coimbatore

2023–2027

CGPA: 7.91/10

Higher Secondary Education

Kamala Niketan Montessori School

83%

Experience

Firmware Intern — Grace Motion Controls, Bangalore

Nov 2025 – Dec 2025

Developed EV motion-control optimization workflows using physics-based motor models and GEKKO. Designed firmware-ready algorithms for motor efficiency mapping and dynamic speed profile generation. Built data analysis pipelines processing feedback control signals, torque–speed characteristics, and energy consumption metrics under varying operational loads.

Full Stack Development Intern — VDart, Trichy

Nov 2024 – Dec 2024

Engineered a resume parsing and ATS platform using a Python backend and structured REST APIs. Designed automated data extraction pipelines and validation workflows, improving structured output accuracy by **23%**. Optimized backend processing pipelines for high-volume document analysis, reducing average processing time by **35%**.

Projects

Smart City Traffic Congestion Prediction System

Spark, Kafka, HDFS, GBT

Designed an end-to-end traffic prediction system using distributed big data. Processed **46M+ GPS trip records** with Apache Spark and Hadoop HDFS; engineered spatial grid features (500m cells) and temporal lags (15/30/60 min). Developed Gradient Boosted Trees (One-Vs-Rest) classifying congestion levels with **79% test accuracy** and weighted F1 **0.76**, outperforming Logistic Regression baseline (**64%**). Built real-time streaming pipeline with Kafka and Spark Structured Streaming handling **1,000+ events/sec** at **3-second latency**. Optimized storage: **120GB raw CSV → 28GB Parquet** with Snappy (**77% reduction**).

VisualVerse: Multi-Modal NLP System for Visual Knowledge Extraction

spaCy, scikit-learn, FastAPI, React

Built an end-to-end NLP system converting unstructured text into mind maps and comic-style visual narratives. Trained on **7,000 documents** (BBC News, short stories). Narrative classification: Random Forest achieved **91% accuracy**, improving over Naïve Bayes baseline (**74%**). Keyphrase extraction with Gradient Boosting reached **Precision@5 = 0.68**, surpassing TF-IDF (**0.42**). LDA topic clustering (15 topics, coherence **0.52**) enabled automatic story segmentation. User study (n=25) rated comprehension improvement **4.3/5** vs raw text. Deployed with FastAPI backend and React + D3.js frontend.

ChronosAI — Hybrid Time-Series Ensemble for Productivity Forecasting

LSTM, ARIMA, Prophet, SHAP

Developed multi-model forecasting framework combining LSTM, ARIMA(1,1,1), and Prophet with inverse-error weighted ensemble. Reduced forecasting error to **62% MAPE** from individual model errors (Prophet: 84%, LSTM: 71%). Processed high-frequency behavioral data sampled every 10 seconds (**8,600+ data points/day per user**) and built a VAR model with Granger causality identifying mood as a significant predictor of productivity (p < 0.05). Implemented SHAP for explainability; mood accounted for **31%** of feature importance.

Open Notebook — Persistent RAG-Based AI Knowledge System

LangChain, FAISS, SurrealDB, FastAPI

Architected a persistent AI knowledge system using chunking (512 tokens, 128 overlap), OpenAI embeddings (1536-dim), and FAISS vector search. Retrieval Hit@5 improved from **0.53** (keyword search) to **0.87**; answer relevance from **3.1/5** to **4.4/5**. Knowledge graph extraction using spaCy + custom relation extraction achieved precision **0.72**, recall **0.68**. Deployed with FastAPI orchestration, LangChain/LangGraph agents, WebSocket task execution, and Docker containers.

Honeypot Security Dashboard — Real-Time Threat Monitoring

React, Flask, scikit-learn, Leaflet

Built a full-stack honeypot platform with 6 decoy endpoints (fake login, API, admin panel) capturing **2,300+ malicious requests** over 2 weeks. Attack classification: regex alone caught **63%**; adding TF-IDF + Random Forest improved to **89% accuracy** across SQLi (n=412), XSS (n=287), command injection (n=156), and reconnaissance (n=1,445). Real-time dashboard with Server-Sent Events streaming — average detection time: **4.2 seconds**. GeoIP-based attack mapping using Leaflet.

Small Vessel Detection from SAR Imagery using Deep Learning

PyTorch, Detectron2, ResNet, OpenCV

Fine-tuned on SAR Ship Detection Dataset (**43,000 images**). ResNet-50-FPN achieved **0.73 mAP@0.5** on inshore scenes; ResNet-18 achieved **0.68 mAP** with **2.3× faster inference** (42ms vs 98ms). Model selection logic switches based on scene complexity: combined system maintained **0.71 mAP** with **51ms average inference**, outperforming baseline Faster R-CNN (no FPN) at **0.61 mAP**. Designed for real-time maritime surveillance.

Namma Vayal — AI-Powered Agricultural Marketplace

Flutter, TFLite, RAG, LangChain

Engineered Flutter app with TFLite disease diagnosis — fine-tuned EfficientNet-Lite0 on PlantVillage (**54k images, 33 classes**) achieving **84% accuracy, 6MB model size, 120ms inference** on mid-range Android. Implemented RAG for farmer queries: crawled Tamil Nadu Horticulture Department, IMD weather APIs, eNAM market prices, and government scheme documents into Chroma vector DB (**1,200+ documents**). Retrieval relevance improved from **58%** (keyword search) to **87%** with hybrid search.

(n=50 queries). Architected direct farmer-to-buyer marketplace with 12-hour bidding windows and UPI integration. Finalist, Gen AI CBE Hackathon 2025.

Doc-Patient Connect — Full-Stack Patient–Doctor Platform

Next.js, Prisma, PostgreSQL, Vonage

Engineered production-grade telehealth platform with appointment scheduling, real-time video consultations, and secure medical record access. Integrated Clerk for authentication and Vonage API for low-latency video sessions — average connection time **12 seconds**. Implemented role-based access control (patient/doctor/admin) with Prisma and PostgreSQL; live notifications via WebSockets. Deployed on Vercel with **99.9% uptime** during testing.

Enterprise Forecasting and Scenario Planning Platform

Python, SQL, ETL, Time-Series

Designed data analytics system to forecast weekly operational metrics and evaluate planning scenarios. Built automated ETL pipelines ingesting and processing large historical datasets (**2M+ records**). Implemented time-series forecasting models (Prophet, ARIMA) achieving **18% lower MAPE** than baseline moving average. Developed variance analysis dashboards to compare predicted vs actual performance for operational planning.

Autonomous Business Intelligence Insight Engine

Python, Anomaly Detection

Developed automated analytics system that detects anomalies in business datasets and generates structured insights. Implemented statistical anomaly detection (Z-score, IQR) and Python-based analysis pipelines identifying deviations in **15+ key performance metrics**. Built automated reporting workflows translating analytical outputs into executive-ready insights, reducing manual analysis time by **80%**.

Enterprise Data Warehouse and BI Analytics Stack

PostgreSQL, Data Warehousing, SQL

Architected end-to-end business intelligence pipeline integrating ETL workflows, relational databases, and analytics dashboards. Designed normalized and star-schema database structures handling **10M+ rows** and optimized SQL queries achieving **40% faster analytical workloads**. Built dashboards for KPI tracking, trend analysis, and operational performance monitoring.

DL-Based Visual Detection System for Maritime Surveillance

PyTorch, Detectron2, ResNet

Developed deep learning detection pipeline using ResNet-FPN architectures for SAR satellite imagery analysis. Designed model selection workflows balancing accuracy and inference latency for large-scale imagery processing. Evaluated model performance using COCO metrics; achieved **0.71 mAP** with inference under **60ms** on consumer GPU.

IoT-Based Healthcare Monitoring and Disease Prediction

Arduino, scikit-learn, Streamlit

Developed IoT healthcare system using Arduino Mega 2560 with SpO2, heart rate, and temperature sensors. Implemented ML models (SVM, Logistic Regression) for diabetes, heart disease, and Parkinson's prediction achieving **82% accuracy** across three conditions. Built real-time data visualization dashboards using Streamlit for live patient monitoring.

Robotic Arm Mounted on Four-Wheeled Drive Vehicle

Arduino, C++ , Motion Control

Designed and built a four-wheeled robotic vehicle with a mounted arm for object manipulation. Implemented motion-control algorithms with obstacle detection and pick-and-place automation. Achieved **92% success rate** in controlled pick-and-place operations.

Analysis of Sequence Alignment Methods for Cyanobacteria Strains

Python, Biopython

Compared Needleman–Wunsch and Smith–Waterman alignment algorithms with K-means clustering on gene sequences. Computed similarity scores and performed cluster analysis to infer evolutionary relationships. Developed Python pipelines processing **500+ gene sequences** and visualizing phylogenetic trees.

KNN-Based Stock Price Movement Predictor

Python, yfinance, scikit-learn

Analyzed Apple stock data (2021–2023) using yfinance and built a KNN-based price movement classifier. Implemented data pre-processing, feature engineering (RSI, moving averages), and model evaluation pipelines. Achieved **61% accuracy** in predicting next-day direction, outperforming random baseline (50%).

University Management System

SQL, Database Design

Developed academic management system handling student records, faculty data, course enrollments, and fee structures. Implemented secure database integration with role-based access control. Designed schema for **10+ entities** and optimized queries for reporting.

Urban Utopia — Property Listing Platform

HTML, CSS, JavaScript, React

Built responsive real estate platform with advanced property search, filtering, and interactive listings. Implemented dynamic filtering with **10+ criteria** and map-based visualization.

Handwriting Recognition and Authentication System

Python, TensorFlow, Computer Vision

Built CNN-based handwriting authentication system verifying user identity through handwriting patterns. Implemented template matching and similarity scoring; achieved **94% authentication accuracy** on a dataset of 50 users.

Achievements

Runner-Up, Gen AI CBE Hackathon 2025 — Finalist, Infosys Global Hackathon 2025

Technical Skills

Programming:	Python, SQL, C/C++ , Java, Dart, JavaScript, TypeScript
AI/ML:	PyTorch, TensorFlow, scikit-learn, CNNs (ResNet, EfficientNet), RNNs/LSTM, Computer Vision, NLP, RAG, SHAP, Time-Series (ARIMA, Prophet)
Big Data:	Apache Spark, Hadoop HDFS, Kafka, Spark Streaming
Web & Mobile:	React, Next.js, Flutter, FastAPI, Flask, Node.js, REST APIs, WebSockets
Backend & DB:	PostgreSQL, SurrealDB, Prisma, MongoDB, Firebase, Supabase, SQLAlchemy
Cloud & DevOps:	AWS S3, AWS RDS, Docker, Git, Linux
Embedded & IoT:	Arduino, C++ , Sensor Integration, GEKKO
Visualization:	D3.js, Streamlit, OpenCV, NumPy, Pandas
Languages:	English (Fluent), Tamil (Native), Hindi (Fluent)